

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 40)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

.....

.....

0 1 . 2

Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 1 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

.....

.....

.....

.....

0 1 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

.....

.....

.....

.....

.....

.....

10

0 2

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

0 2 . 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]

Answer

thousand tonnes

0 2 . 2

Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer thousand tonnes

0 2 . 3

Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 4

Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort

.....

.....

.....

Mean fish size and mean age

.....

.....

.....

10

03 . 1

Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems	
Energy source			ox
Use of materials			
Nature of wastes			
Temperature of processes			

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

03 . 2

Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

.....

.....

.....

.....

03 . 3

Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

.....

.....

.....

.....

0 3 . 4

Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

.....

.....

.....

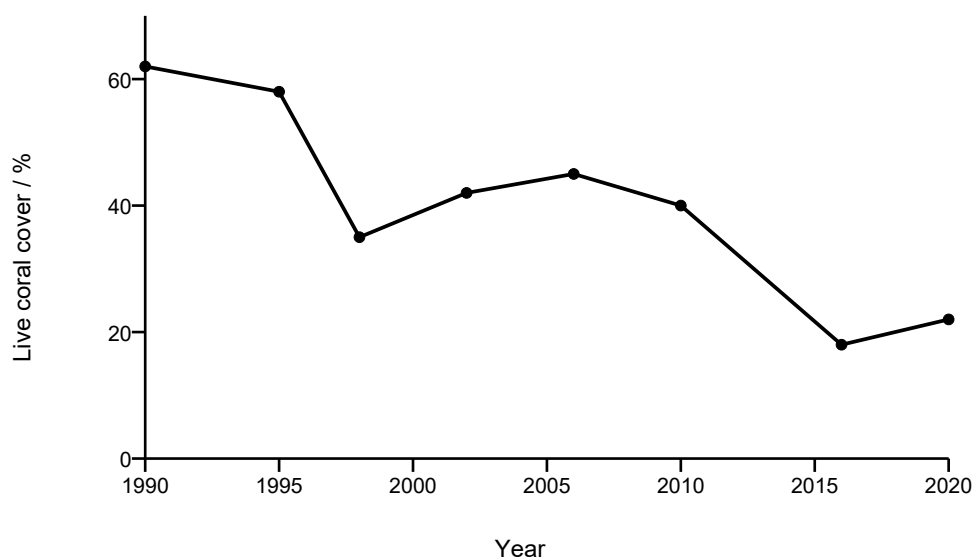
.....

10

0 4

Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 4 . 1

Describe the changes in coral cover shown in Figure 1.

[2 marks]

.....

.....

.....

.....

0 4 . 2

Explain what coral bleaching is and give **three** causes.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 4 . 3

Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 4 . 4

Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

.....

.....

10

0 5

Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 5**1**

Explain what the standard deviation values tell you about the data.

[2 marks]

.....

.....

.....

.....

0 5**2**

Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

.....

.....

.....

.....

0 5**3**

The test gave a probability $p = 0.30$. State what this means and what the student should conclude.

[2 marks]

.....

.....

.....

.....

0 5**4**

Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

.....

.....

.....

.....

05 . 5

Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

.....

.....

.....

.....

10

06 . 1

Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis

☐

B Radio tracking

☐

C Lincoln index (mark-release-recapture)

☐

D Leg rings

☐

06 . 2

Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

.....

.....

.....

.....

06 . 3

Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

.....

.....

.....

.....

5

07

Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha ⁻¹	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

07**1**

Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

07**2**

Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

07 . 3

Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

.....

.....

.....

.....

.....

.....

10

08

The Isle of Eigg (area 31 km², population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

08 . 1

Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

.....

.....

.....

.....

.....

.....

08 . 2

Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

1

2

3

0 8 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

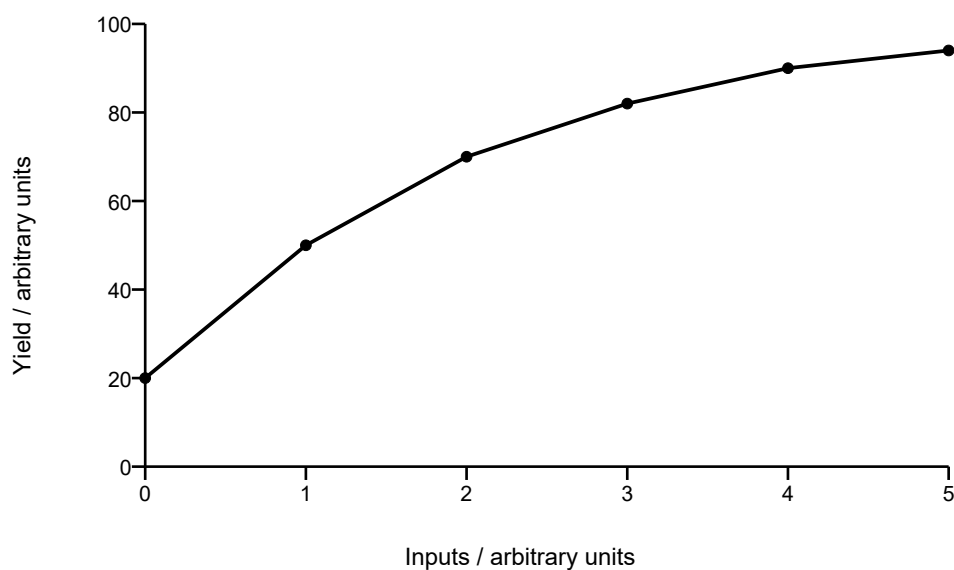
0 8 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

10

0 9

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1**0 9****1**

Describe the relationship shown in Figure 1.

[2 marks]**0 9****2**

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]**Answer**

units

0 9**3**

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

09 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

09 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

10

10 . 1

Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

[illegible]

.....

.....

.....

1	0
---	---

 .

2

 State what is meant by **biodiversity**.

[1 mark]

.....

.....

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

OR

1 1 . 2

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

END OF QUESTIONS